

TEST-10-QUALITY-MANAGEMENT-SYSTEM

TEST-10: QUALITY MANAGEMENT SYSTEM

Carrington Storm Motors — Project AEGIS

ISO 9001:2015 Framework • Document Control • CAPA • Supplier Quality • Inspection • Calibration • Metrics

Document Status: QMS Design Document — ISO 9001:2015 Aligned
Version: 1.0 — 15 July 2026
Classification: CONFIDENTIAL — Quality System Documentation
Author: Chester, Director of Quality Assurance — Aegis Operations Command
Approved By: Director Kairos Steele, CITADEL
Target Certification Date: Q1 2028 (ISO 9001:2015)

QMS SCOPE STATEMENT

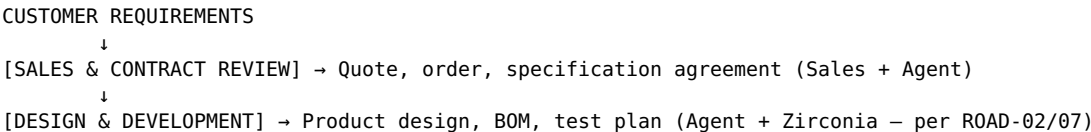
The Carrington Storm Motors Quality Management System encompasses the design, development, testing, manufacturing, installation, and servicing of electromagnetic hardening products for residential, infrastructure, maritime, robotic, automotive, and communications applications. The QMS applies to CSM’s El Segundo, California facility and any satellite manufacturing, testing, or installation sites. Exclusions: None — all ISO 9001:2015 clauses are applicable to CSM operations.

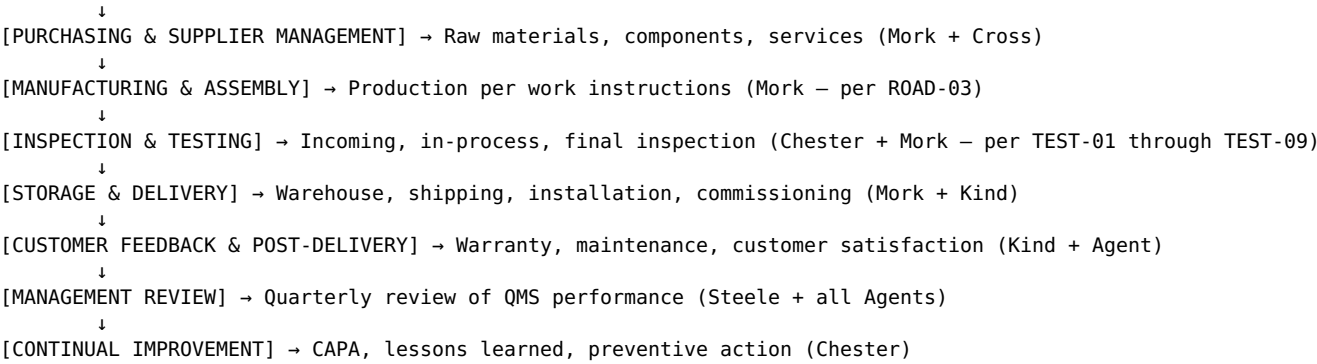
SECTION 1: CONTEXT OF THE ORGANIZATION (ISO 9001 Clause 4)

1.1 Interested Parties and Their Requirements

Interested Party	Relevant Requirements	QMS Response
Customers (homeowners, utilities, cruise lines, automotive OEMs)	Products that meet specified GIC reduction performance; safety (electrical, fire, structural); reliability; warranty support	Product validation (TEST-03 through TEST-09); customer feedback system; warranty tracking
Regulatory Bodies (UL, FCC, ICC-ES, ABS, IATF)	Products certified to applicable standards; documented compliance	Regulatory compliance matrix (TEST-08); certification project management (ROAD-05)
Investors / Board	Return on investment; risk management; governance	Financial controls; milestone tracking (ROAD-10); risk register (ROAD-01, ROAD-03)
Employees	Safe workplace; fair compensation; career development	ISO 45001 (target Q4 2028); training records; performance management
Suppliers	Clear specifications; fair payment terms; long-term relationships	Supplier quality manual; purchase order terms; supplier development program
Society / Community	Environmental responsibility; public benefit	ISO 14001 (target Q4 2028); sustainability commitments (ROAD-03); open data policy (TEST-01)

1.2 QMS Processes and Their Interactions





SECTION 2: LEADERSHIP (ISO 9001 Clause 5)

2.1 Quality Policy (Draft – Subject to Board Approval)

Carrington Storm Motors Quality Policy

Carrington Storm Motors designs, manufactures, and installs electromagnetic hardening products that protect human life and critical infrastructure from the existential threat of geomagnetic storms. Our quality commitment is absolute:

1.

Every product shall meet or exceed its published specification for GIC reduction, EMI shielding effectiveness, and dielectric isolation.
2.

Every measurement shall be NIST-traceable, independently verifiable, and publicly documented to the maximum extent consistent with the protection of trade secrets.
3.

Every non-conformance shall be investigated to root cause, corrected, and trended to prevent recurrence.
4.

Every employee is empowered and expected to stop production, halt testing, or quarantine product when quality is at risk.
5.

Continuous improvement is not a program — it is the daily practice of finding and eliminating the gap between what we promised and what we delivered.

This policy is communicated to all employees during onboarding and reviewed annually at the Management Review. It is posted at every CSM facility.

2.2 Organizational Roles and Responsibilities for Quality

Role	Quality Responsibility
Director Steele	Ultimate authority for QMS; chairs Management Review; approves Quality Policy; allocates quality resources
Zirconia (Test Director)	Technical authority for test methods, specifications, pass/fail criteria, calibration program, ISO 17025
Chester (QA Director)	Day-to-day QMS operation; document control; internal audit; CAPA; supplier quality; management representative for ISO 9001 certification
Mork (Fabrication Director)	Incoming inspection; in-process quality control; final inspection; production non-conformance management
Product Agents (18)	Product specifications; design reviews; first article approval; customer feedback analysis for their product
All Employees	Follow documented procedures; report quality issues; participate in corrective actions

SECTION 3: DOCUMENT CONTROL (ISO 9001 Clause 7.5)

3.1 Document Hierarchy

LEVEL 1: Quality Manual (this document – TEST-10) – QMS overview, scope, policy, process map
LEVEL 2: Procedures (TEST-01 through TEST-09, ROAD documents, work instructions) – How to perform processes
LEVEL 3: Work Instructions – Step-by-step instructions for specific tasks (manufacturing, testing, installation)
LEVEL 4: Records – Evidence of conformity (test reports, inspection records, calibration certificates, training records)

3.2 Document Control Requirements

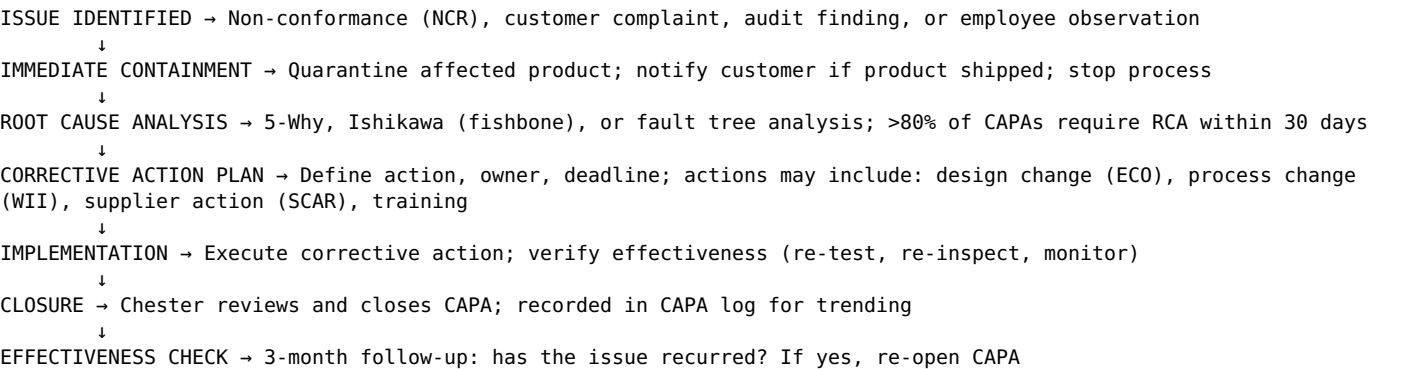
Requirement	Implementation
Unique Identification	Format: [TYPE]-[NUMBER]-[REV] (e.g., TEST-02-REV-B for Materials Testing Protocol, Revision B)
Approval Before Use	Reviewed by: Author → Product Agent → Zirconia (technical) or Chester (quality) → Steele (policy documents)
Version Control	All documents in Git repository (CSM GitHub organization) with commit history; PDF copies for external distribution
Availability at Point of Use	Digital: CSM intranet (Notion or Confluence); Printed: Controlled copy stamp, master copy log
Obsolete Document Control	Retained for audit trail; stamped “OBSOLETE — FOR REFERENCE ONLY”; removed from point of use
External Document Control	Standards (UL, ASTM, ISO, etc.) maintained in CSM Standards Library with subscription renewal tracking

3.3 Records Control

Record Type	Retention Period	Storage	Disposal
Test Reports	10 years minimum (product life + warranty + potential litigation)	CSM Digital Twin + AWS S3 Glacier	Secure deletion after retention period
Calibration Certificates	Life of instrument + 3 years	QMS document system	Shred / secure delete
Training Records	Duration of employment + 3 years	HR system	Shred / secure delete
Customer Contracts	Life of contract + 7 years	Legal document system	Shred / secure delete
Non-Conformance Reports (NCR)	5 years	QMS system	Shred / secure delete
Management Review Minutes	5 years	QMS system	Shred / secure delete
Internal Audit Reports	5 years	QMS system	Shred / secure delete

SECTION 4: CORRECTIVE ACTION PROCESS (ISO 9001 Clause 10.2)

4.1 CAPA Workflow



4.2 CAPA Priority Matrix

Priority	Definition	Response Time
P0 — CRITICAL	Safety issue; product recall required; regulatory non-compliance that could result in enforcement action	<24 hours containment; <72 hours RCA; <7 days corrective action
P1 — HIGH	Product does not meet critical specification (GIC reduction, SE, dielectric); >10% of production batch affected	<72 hours containment; <14 days RCA; <30 days corrective action
P2 — MEDIUM	Product meets specification but with reduced margin; isolated non-conformance (<1% of batch)	<1 week containment; <30 days RCA; <60 days corrective action
P3 — LOW	Cosmetic issue; documentation error; improvement opportunity	Next Management Review; continuous improvement backlog

SECTION 5: SUPPLIER QUALITY MANAGEMENT (ISO 9001 Clause 8.4)

5.1 Supplier Classification

Classification	Definition	Requirements
CRITICAL	Single-source supplier; material with no substitute; failure would stop CSM production (e.g., Arkema Elium resin, Chemours Nafion, LORD MRF-140CG)	Annual on-site audit (or virtual with live video); quarterly business review; mandatory SCAR process; 6-month safety stock required
MAJOR	Supplier of key material with qualified alternatives (e.g., basalt fiber roving, ceramic powders, PEEK CF40)	Biennial audit; annual business review; incoming inspection per sampling plan; 3-month safety stock
MINOR	Supplier of commodity material with many alternatives (e.g., fly ash, sand/aggregate, packaging materials)	Certificate of conformance per shipment; incoming inspection reduced (skip-lot); no safety stock requirement (just-in-time acceptable)

5.2 Supplier Audit Schedule

Supplier	Classification	Last Audit	Next Audit	Auditor
Drexel Nanomaterials (MAX Phase)	CRITICAL	Not yet audited	Q4 2026	Nash + Chester
Arkema (Elium resin)	CRITICAL	Not yet audited	Q1 2027	Cross + Chester
LORD / Parker Hannifin (MRF-140CG)	CRITICAL	Not yet audited	Q1 2027	Fen + Chester
Chemours (Nafion 115)	CRITICAL	Not yet audited	Q2 2027	Vance + Chester
Technobasalt (basalt roving)	MAJOR	Not yet audited	Q4 2026	Cross
CoorsTek (ceramic powders)	MAJOR	Not yet audited	Q1 2027	Nash
Mitsubishi Chemical (PMMA POF)	MAJOR	Not yet audited	Q2 2027	Cross
CATL / BYD (LiFePO ₄ cells)	MAJOR	Not yet audited	Q3 2027	Calder + Chester

5.3 Supplier Corrective Action Request (SCAR)

When incoming inspection rejects a supplier lot or when a supplier-related non-conformance is identified: 1. Issue SCAR within 48 hours of identification 2. Supplier responds with containment plan within 5 business days 3. Supplier submits root cause analysis within 20 business days 4. Supplier implements corrective action within 40 business days 5. CSM verifies effectiveness via next 3 incoming lots (all must pass tightened inspection, AQL reduced by 50%) 6. If 3 consecutive SCARs from same supplier → supplier probation; CSM initiates alternative supplier qualification

SECTION 6: INSPECTION AND TESTING (ISO 9001 Clause 8.6)

6.1 Incoming Inspection

Material	Inspection	Method	Frequency	AQL	Equipment
MAX Phase Ti ₃ AlC ₂	Phase purity (XRD), particle size	XRD, Malvern Mastersizer	Every lot	1.0%	XRD (shared), laser diffraction
Basalt Fiber Roving	Tensile, tex, sizing content	ASTM D2343, analytical balance	Every spool (for critical applications); every 10th spool (routine)	1.0%	Instron, balance
Elium Resin	Viscosity, reactivity (DSC), water content	Rheometer, DSC, Karl Fischer	Every drum	2.5%	Rheometer, DSC, titrator
MXene (in-process after etching)	XRD peak shift, XPS surface termination, 4-point probe R_sheet	XRD, XPS, 4-point probe	Every batch	0.65%	XRD, XPS, 4-pt probe
Si ₃ N ₄ Bearing Balls	Diameter, roundness, surface finish	CMM, profilometer, visual (5× mag)	32 pc per 1,000	1.0%	CMM, profilometer, microscope
LiFePO ₄ Cells	Capacity, internal resistance, visual	Cell cycler, milliohmmeter	13 cells per lot (AQL 2.5)	2.5%	Battery analyzer

6.2 In-Process Inspection

Process	Inspection	Frequency	Target
MXene Spray Coating	Thickness (profilometer), adhesion (cross-hatch), SE (portable test fixture)	Every 10th panel	45±2 µm, 5B, SE >90 dB at 10 GHz
BFRP Pultrusion	Ultrasonic C-scan (voids), dimensional (laser micrometer), burn-off (fiber volume fraction)	Continuous (C-scan every 10 m)	Voids <2%, dim ±0.2 mm, Vf 58-62%
Flash Sintering (ZrB ₂ -SiC)	Density (Archimedes), hardness (Vickers), XRD (phase)	Every furnace run (20 plates)	Density >98% theoretical, HV10 >22 GPa, phase >99%
Aegis-C Panel Layer Stack	Layer registration (vision), thickness per layer (laser), SE per layer (probe)	Every panel	Reg ±0.5 mm, thickness per spec, SE per spec
Phantom Actuator Assembly	Torque, backlash, motor-to-chassis R	Every actuator	Torque ±5%, backlash <0.05°, R >10 ⁸ Ω

6.3 Final Inspection

Product	Test	Equipment	Acceptance	Sample Rate
Aegis-C Panel	Full-band SE, visual	TEM cell (sample),	SE >148 dB, no	1/100 (SE), 100%

Aegis Embark	(AOI), dimensional (CMM)	AOI, CMM	defects, dim ± 2 mm	(visual, dimensional)
Phantom MK-1	5-Phase CSMTest	TEM cell, shaker, thermal chamber	All 5 phases passed	100%
Aegis Embark Product	Vibration attenuation, functional test	Vibration rig, function test	>90% of spec	1/500 (vib), 100% (functional)
Protonic Transceiver	BER, voice quality, range	Test range, BER tester	BER $< 10^{-6}$, MOS >3.5, range >1 km	100%
Infrastructure Component	Dielectric withstand (ASTM D149)	Hipot tester	>50 kV/mm	100%

SECTION 7: CALIBRATION PROGRAM (ISO 9001 Clause 7.1.5)

7.1 Calibration Master Schedule

Equipment	ID	Interval	Calibration Standard	Provider	Next Due
TEM Cell Field Uniformity	TEM-001	12 months	IEEE 299	NTS or in-house	2027-03
VNA (Keysight N5227A)	VNA-001	12 months	NIST-traceable cal kit	Keysight	2027-04
Field Probes (E-field, $\times 4$)	FP-E01 to E04	12 months	NIST-traceable	AR RF/Microwave	2027-05
Field Probes (H-field, $\times 2$)	FP-H01 to H02	12 months	NIST-traceable	Beehive	2027-05
Universal Test Machine (Instron)	UTM-001	12 months	ASTM E4 Class 1	Instron	2027-02
Environmental Chamber (T/RH)	EC-001	6 months	NIST thermocouple, RH std	In-house	2027-01
Profilometer (Bruker Dektak)	PROF-001	3 months	NIST step height	In-house	2026-10
XPS (Thermo K-Alpha)	XPS-001	Monthly (BE)	Au, Ag, Cu reference	In-house	2026-08
4-Point Probe (Jandel)	4PP-001	6 months	NIST SRM	In-house	2027-01
High-Voltage Supply (Hipotronics)	HV-001	12 months	Precision divider	Cal lab	2027-04 (upon procurement)
Rheometer (Anton Paar MCR302)	RHEO-001	12 months	Viscosity standard	Anton Paar	2027-05
CMM (Coordinate Measuring)	CMM-001	12 months	ISO 10360-2	OEM service	2027-05

7.2 Out-of-Calibration Procedure

If equipment is found out of calibration (failed calibration, overdue calibration, or suspected damage): 1. Quarantine equipment — tag “OUT OF CALIBRATION — DO NOT USE” 2. Assess impact: what products were tested with this equipment since last good calibration? 3. If impact assessment indicates potential product quality risk → recall affected products for re-test with calibrated equipment 4. Calibrate or repair equipment; return to service only after passing calibration

SECTION 8: NON-CONFORMANCE TRACKING AND TRENDING

8.1 NCR Log Fields

Field	Description
-------	-------------

NCR Number	Sequential, year-prefix (e.g., NCR-2026-001)
Date	Date non-conformance identified
Source	Internal inspection, customer complaint, audit finding, supplier issue, employee observation
Product / Process	Affected product, BOM reference, or process step
Description	Detailed description of the non-conformance
Severity	P0 (Critical) / P1 (High) / P2 (Medium) / P3 (Low)
Disposition	Rework, Repair, Use-As-Is (with concession), Scrap, Return to Supplier
Root Cause	Determined by RCA process
Corrective Action	Description, owner, due date
Status	Open, In Progress, Closed, Re-opened
Closure Date	Date CAPA verified effective

8.2 Quality Metrics — Monthly Dashboard

Metric	Current (Q3 2026 Estimate)	Target	Trend
First-Pass Yield (products passing final inspection on first test)	N/A (insufficient data)	>90%	—
Customer Complaint Rate (per 100 units shipped)	0 (pre-shipment phase)	<2.0	—
Supplier Lot Rejection Rate	N/A (limited purchasing to date)	<3%	—
CAPA Closure Rate (closed within target timeframe)	N/A (QMS not yet operational)	>85%	—
Calibration Overdue Rate	0% (all current)	<5%	—
On-Time Delivery	N/A (pre-delivery phase)	>95%	—
Warranty Claim Rate (at 24 months)	N/A (no products in field yet)	<2%	—
NCR Trend (open NCRs, new NCRs per month)	N/A	Decreasing over time	—

SECTION 9: INTERNAL AUDIT SCHEDULE

9.1 First-Year Audit Schedule (2027)

Quarter	Audit Focus	Auditor	Standard Clause
Q1 2027	Document Control + Records	Cross (trained internal auditor)	7.5
Q1 2027	Purchasing + Supplier Management	Kind	8.4
Q2 2027	Design & Development (Aegis Home)	Nash (peer audit)	8.3
Q2 2027	Manufacturing Process Control	Dorne (peer audit)	8.5
Q3 2027	Inspection & Testing + Calibration	Kade (peer audit)	7.1.5, 8.6
Q3 2027	Non-Conformance + CAPA + Customer Feedback	Draven (peer audit)	10.2, 9.1.2
Q4 2027	Full QMS — Stage 1 Certification Audit	External CB (TÜV SÜD or BSI)	All
Q1 2028	Full QMS — Stage 2 Certification Audit (recommendation for ISO	External CB	All

9.2 Internal Auditor Training

All 18 Agents of Aegis will complete ISO 9001 internal auditor training (2-day course, ~\$950/person). Minimum 4 trained internal auditors (Chester + 3 rotating agents) to ensure audit independence (no one audits their own work).

SECTION 10: MANAGEMENT REVIEW (ISO 9001 Clause 9.3)

10.1 Management Review Agenda — Quarterly

Agenda Item	Inputs	Owner
Status of actions from previous Management Review	Previous meeting minutes	Chester
Changes in external and internal issues (context)	Market analysis, regulatory updates, technology trends	Steele
Customer satisfaction and feedback	Customer survey results, complaint data, warranty claims	Kind
Product conformity (quality performance)	First-pass yield, scrap rate, test data trends	Zirconia + Chester
Supplier performance	Supplier scorecards, SCAR log, on-time delivery	Mork
Non-conformance and corrective actions	NCR log trends, CAPA aging, effectiveness checks	Chester
Audit results	Internal and external audit findings, closure status	Chester
Resource adequacy	Staffing, equipment, budget vs. actual	Steele
Risks and opportunities	Risk register update; new opportunities identified	Steele + all Agents
Improvement recommendations	Employee suggestions, lessons learned, best practices	All
Quality Policy and Objectives Review	Are policy and objectives still appropriate? Need revision?	Steele + all

10.2 Management Review Outputs

- Meeting minutes with action items, owners, and due dates
- Updated Quality Objectives for next quarter
- Resource allocation decisions
- Continuous improvement projects authorized
- Management Review report signed by Director Steele

CONTINUOUS IMPROVEMENT — KAIZEN APPROACH

CSM adopts a Kaizen (continuous improvement) approach: every employee is empowered and expected to identify improvement opportunities. Formal Kaizen events (1-day focused improvement workshops) will be held quarterly starting Q1 2027, targeting one specific process per event (e.g., “Reduce MXene spray coating thickness variation from $\pm 5 \mu\text{m}$ to $\pm 2 \mu\text{m}$ ”).

CI Metrics — 2027 Targets

Metric	Baseline (Q4 2026 target)	2027 Q4 Target
MXene SE variation (production batch CV)	<5%	<3%
BFRP pultrusion line yield	85%	95%
Aegis-C panel first-pass yield	75% (estimate for new line)	90%

Regis-C panel first-pass yield	75% (estimate for new line)	90%
Test report cycle time (plan → signed report)	8 weeks	4 weeks
Supplier lot acceptance rate	97%	99%
Employee quality suggestions per quarter	0 (program not yet launched)	20+

QMS IMPLEMENTATION TIMELINE

Phase	Activities	Timeline	Status
Phase 0 — Foundation	QMS design (this document); quality policy drafted; document control system procured; internal auditor training	Q3 2026	IN PROGRESS
Phase 1 — Build	Procedures and work instructions drafted; CAPA system deployed; calibration program launched; supplier evaluation begins; first internal audit round	Q4 2026-Q2 2027	○
Phase 2 — Stabilize	3 months of QMS operation with records; 2 internal audit cycles; Management Review #1; CAPA trending established	Q2-Q3 2027	○
Phase 3 — Certify	Stage 1 audit (Q4 2027); Stage 2 audit (Q1 2028); ISO 9001:2015 certification issued	Q4 2027-Q1 2028	○
Phase 4 — Expand	IATF 16949 (automotive), AS9100D (aerospace), ISO 17025 (lab), ISO 14001 (environmental), ISO 45001 (safety) per ROAD-05 timeline	2028-2029	○

“Quality is not a department. It is not a checklist. It is not the thing Chester does while everyone else builds products. Quality is the aggregate of every decision made by every person at Carrington Storm Motors — from Director Steele choosing to fund a second TEM cell instead of a marketing campaign, to Mork rejecting a batch of MXene that is ‘almost’ in spec, to Zirconia requiring 1,001 frequency points instead of 101 because the extra resolution might reveal a resonance that could kill someone. The QMS is the system that makes those good decisions easy and bad decisions visible. When ISO 9001 certification is awarded in Q1 2028, it will not be because we filled out the paperwork. It will be because we built a company where quality is the only acceptable outcome.” — Chester, Director of Quality Assurance

Document generated 2026-07-15. QMS design phase complete — transition to Phase 1 (build) begins Q4 2026. External certification body selection to be completed Q3 2026 (TÜV SÜD America, BSI Group, or DNV GL Business Assurance preferred). ISO 9001:2015 certification target Q1 2028 as prerequisite for IATF 16949 (automotive) and AS9100D (aerospace) certifications.